MDPI

Review

Data-Efficient Real-World Robotic Manipulation: A Critical Review of Data Collection, Augmentation, Adaptation, and Generalization

Xiongyule Zhou

School of Mechanical, Electronic and Control Engineering, Beijing Jiaotong University, Beijing 100044, China; jensen.zhou21@gmail.com

Abstract

Real-world manipulation data consume robot access, human attention, setup effort, and recovery time. Methods that reduce target demonstrations may transfer these costs to pretraining, synthetic generation, or online interaction. This structured critical review examines data collection, curation, augmentation, policy learning, adaptation, and physical generalization through a two-stage audit of publicly accessible literature searched through 16 September 2026, emphasizing 2022–2026. Studies are organized by the resource they substitute, the assumptions enabling substitution, and the physical evidence supporting deployment. The synthesis identifies three recurring distinctions: visual diversity versus action coverage, semantic transfer versus contact competence, and demonstration savings versus total system efficiency. Portable collection interfaces trade robot occupancy for reconstruction and feasibility checks; synthetic data trade repeated demonstration for validity assumptions; pretrained and interactive policies trade target supervision for prior investment or additional robot trials. We develop separate marginal-user and allocated-lifecycle ledgers, seven conditional comparisons between competing investments, and a diagnosis-led validation protocol. The protocol couples matched-budget physical retesting with explicit holdouts, intervention accounting, and recovery measurement. The framework complements model- and dataset-centered surveys by making integration, physical validation, and sustained-operation costs explicit.

Keywords: robotic manipulation; data efficiency; learning from demonstration; data curation; synthetic data; imitation learning; vision-language-action models; policy adaptation; generalization; real-world deployment

1. Introduction

Learning-based manipulation has moved from narrowly trained visuomotor skills toward policies that accept language, control different robots, and adapt to new environments. A laboratory must still decide how to allocate resources among demonstrations, prior data, augmentation, and interaction. Demonstration collection requires robot or sensing access, feasible initial states, skilled operators, synchronized observations and actions, and recovery after failure. A recorded trajectory captures only part of this expenditure. Dataset initiatives such as BridgeData V2 and DROID make both the promise of diverse experience and the effort required to obtain it visible [1,2].

Several developments make this question timely. Action chunking and diffusion-based policies improve the use of task-specific demonstrations [3,4]. Portable interfaces

Received:
Revised:
Accepted:
Published:

Copyright: © 2026 by the author. Submitted to *Appl. Sci.* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

change where demonstrations can be collected [5]. Cross-embodiment datasets and generalist policies allow downstream users to reuse knowledge rather than acquire every behavior from the beginning [6–8]. Augmentation can broaden the observations or trajectories associated with a small set of source demonstrations [9–11]. Interactive refinement can improve behavior beyond a fixed demonstration distribution [12,13]. These interventions address different resource constraints; their sample-efficiency claims therefore require distinct accounting boundaries.

This review addresses real-world manipulation by robot arms, grippers, and dexterous hands, including manipulation performed by mobile platforms. Navigation enters only when it changes the manipulation data or deployment problem. Simulation studies are included when they explain an important mechanism or evaluate a benchmark used in this literature; their results are explicitly separated from physical evidence. We emphasize work first disseminated during 2022–2026 and include earlier foundations where needed to explain distribution shift and learning from prior demonstrations [14–16].

1.1. Relationship to Existing Reviews

Jiang et al. explicitly frame robot skill learning around a data dilemma; their publisher abstract identifies simulation–reality combination, state representations, and knowledge sharing as remedies [17]. We use that accessible abstract only to establish topical overlap, not for technical or performance extraction. Foundation-model reviews organize model capabilities and their roles in robot systems [18]. Human-video and diffusion reviews examine particular sources of prior knowledge or policy families [19,20]. Reviews of efficient vision-language-action (VLA) models already cover computational efficiency, training, and data collection [21,22]. Real-world reinforcement learning and force-aware policy reviews address important parts of deployment and physical interaction [23,24].

Recent VLA reviews distinguish components, low-level policies, and high-level planning [25], or organize fusion architectures, datasets, and evaluation while explicitly discussing resource efficiency [26]. Bai et al. already connect manipulation methods to data collection, utilization, and generalization bottlenecks [27]. Parnada et al. explicitly connect contact-rich RL sample efficiency to implementation cost and safety [28]; Acs and Zhong map long-horizon RL and propose reporting recommendations [29]. Sarowar's 2026 preprint on datasets versus robust frameworks also connects data, models, and evaluation [30]. This overlap makes resource accounting, rather than topic coverage alone, the basis for differentiation. Tsuji et al. additionally connect teaching interfaces, sensory modalities, imitation methods, and contact-rich applications [31]. We build on these reviews by tracing *where each saving is booked and which obligations remain outside that boundary*, across task-specific imitation, foundation-policy adaptation, and interactive refinement. STAR-Gen supplies an existing visual, semantic, and behavioral generalization taxonomy [32]; here those shifts determine what must be withheld during physical validation. Table 1 compares the organizing emphases of the full-text-inspected reviews. Our distinction is the joint use of resource substitution, remaining deployment obligations, separate ledgers, and matched-budget physical validation across these method families; efficiency and deployment are already recognized concerns in the preceding literature.

Table 1. Scope-emphasis matrix for overlapping reviews. Y: a central organizing topic; P: supporting discussion. A dash means no dedicated emphasis assigned in this extraction, not absence of the topic. Parenthetical (P) marks a cited preprint. Coll.: collection; Cur.: curation; Aug.: augmentation; Gen.: generalization; Phys.: physical evidence/evaluation; Cost: resource/cost discussion, not necessarily a complete ledger. IL: imitation learning; RL: reinforcement learning; VLA: vision-language-action. Codes describe full-text-inspected versions and are not quality scores. The present review connects these eight areas through resource substitution, separate ledgers, and matched-budget validation.

Review	Coll.	Cur.	Aug.	IL/RL	VLA	Gen.	Phys.	Cost	Distinctive emphasis
Li et al. [18]	P	–	P	Y	Y	Y	P	–	Foundation-model roles
Yu et al. [21] (P)	Y	P	P	P	Y	P	P	P	Efficient VLA lifecycle
Guan et al. [22] (P)	P	P	P	P	Y	P	P	P	VLA computation
Ma et al. [19] (P)	Y	P	P	Y	P	P	P	–	Human-video transfer
Wolf et al. [20]	P	–	Y	Y	P	P	P	–	Diffusion mechanisms
Tang et al. [23]	P	–	P	Y	–	P	Y	P	Physical RL practice
Xie/Correll [24] (P)	Y	–	P	Y	P	P	Y	P	Force-aware control
Hou et al. [33] (P)	P	–	Y	Y	P	Y	P	–	World models
Bai et al. [27] (P)	Y	P	Y	Y	Y	Y	P	P	Manipulation bottlenecks
Parnada et al. [28]	P	–	P	Y	–	P	Y	Y	Contact-RL cost
Acs/Zhong [29]	P	–	P	Y	P	Y	Y	P	Long-horizon RL
Sarowar [30] (P)	Y	P	P	Y	Y	Y	Y	P	Data/model/evaluation
Tsuji et al. [31]	Y	P	P	Y	P	P	Y	P	Contact-rich imitation
Ma et al. [25]	P	P	P	Y	Y	P	P	P	Components/policies/planners
Ud Din et al. [26]	P	P	P	Y	Y	Y	Y	P	Fusion/data/evaluation

1.2. Questions and Contributions

Four questions organize the synthesis: which resource is saved; which assumptions permit the substitution; which generalization is tested on hardware; and whether savings persist after integration, evaluation, intervention, and recovery.

This review develops a resource-aware framework for interpreting and comparing data-efficient real-world manipulation methods through four connected contributions:

1. A resource-substitution taxonomy linking acquisition, curation, augmentation, learning and deployment to the information each intervention supplies.
2. Separate marginal-user and allocated-lifecycle ledgers that preserve the units of human time, robot time, computation, demonstrations, and physical tests.
3. Seven conditional comparisons that identify hidden investments and experiments needed to resolve competing data-efficiency claims.
4. A deployment-oriented protocol that connects failure diagnosis to matched-budget physical validation, untouched test conditions, and recovery accounting.

Figure 1 presents the lifecycle.

2. Review Methodology

2.1. Search Scope and Sources

This structured critical review followed two stages, with a final search cutoff of 16 September 2026 and emphasis on 2022–2026. Stage 1 comprised broad discovery across fourteen query families: manipulation data efficiency, demonstration efficiency, collection, curation, synthetic data, imitation learning, diffusion policies, VLA adaptation, cross-embodiment learning, real-world RL, interactive imitation, failure/recovery supervision, generalization, and physical deployment evaluation. Native searches returned records from Google Scholar, OpenAlex, Crossref, and SpringerLink. Public web-index searches additionally covered arXiv and publisher/conference domains, including IEEE Xplore, ACM Digital Library, ScienceDirect, RSS, CoRL, ICRA/IROS, CVPR, NeurIPS, ICML, and ICLR. Domain-restricted web retrieval is distinguished from native database retrieval in

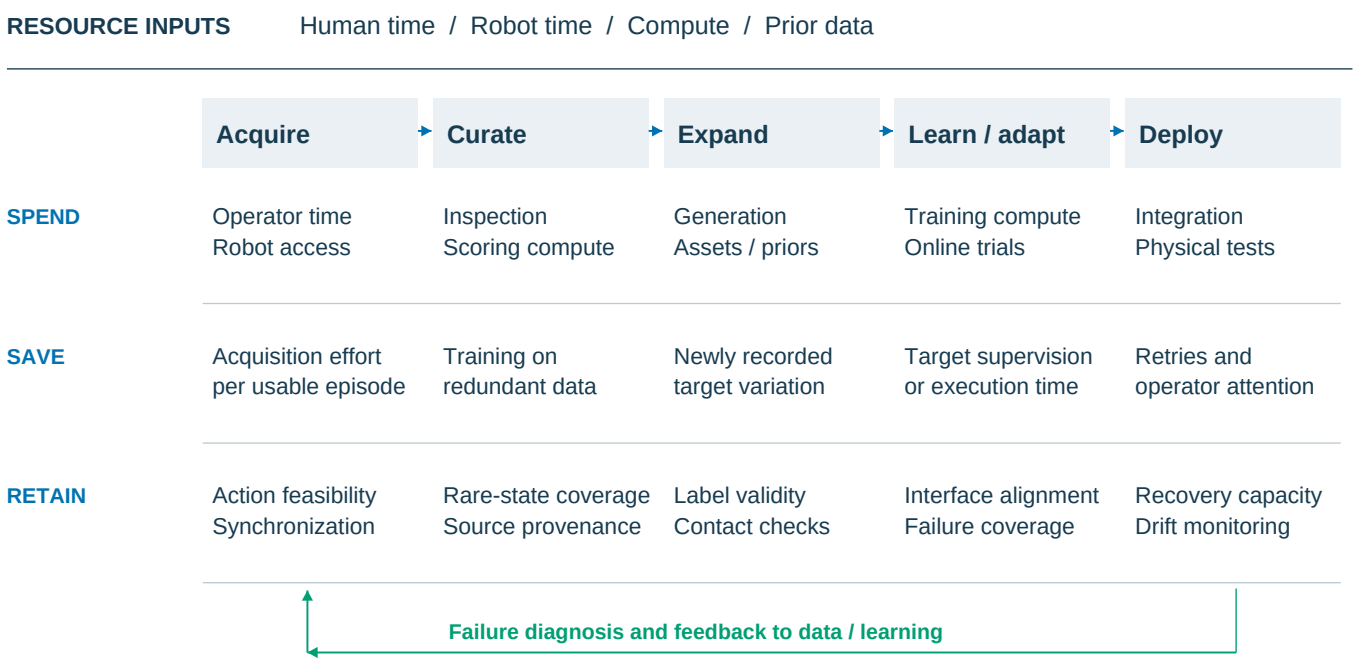


Figure 1. Resource substitution across the manipulation lifecycle. Each stage pairs expenditure with a potential saving and a residual obligation. Savings are conditional mechanisms, not measured effect sizes. Deployment failures feed back to acquisition and learning; prior data and computation remain visible alongside human and robot time.

the search log. Semantic Scholar requests were rate-limited; Scopus and Web of Science did not provide usable search results and were not treated as searched databases.

Stage 2 used gap-directed searches, backward reference inspection, forward citation searches, and exact-title verification. Forward tracing used OpenAlex records for the Universal Manipulation Interface and mutual-information curation papers; reference inspection linked reviews back to primary studies. DOI, repository and official proceedings records resolved bibliographic identity and publication versions. Query-level records, access failures, retrieved candidates, deduplication links, and extraction locators are supplied in Supplementary Files S2–S3. Retrieval was bounded by recorded page limits and purposive relevance selection; these logs enable the procedure to be repeated but do not establish exhaustive coverage or stable search rankings. Counts denote saved records, not estimated search-engine hit totals; unavailable counts are marked NR.

2.2. Eligibility, Deduplication, and Screening

An eligible primary study had to describe a manipulation learning method, dataset, collection interface, or evaluation protocol relevant to the use of limited physical robot data. The central synthesis prioritized physical experiments with identifiable data sources and evaluation conditions. Simulation-only work was retained when it established a mechanism used by physical systems or a widely used evaluation context. Relevant surveys were retained for scope comparison, not substituted for primary evidence of method performance. Earlier works were included selectively for conceptual foundations. English-language full texts were required for detailed technical extraction. An inaccessible review could be retained for a narrowly attributed abstract-level scope statement, but was not coded in the detailed comparison matrix.

We excluded unrelated control, navigation-only, and locomotion-only studies, news articles, repository lists without an associated paper, and records whose title or identity did not match the intended study. Search snippets were discovery aids rather than ev-

idence for quantitative claims. Candidate records were deduplicated by DOI or arXiv identifier, followed by title comparison; conference and journal versions were linked as one study instead of counted as independent experimental evidence. A title–identifier mismatch was detected during verification and excluded before the correct record was retrieved. The final screening register links retained records to their discovery route, verification source, extraction scope, and synthesis use. Relevant candidates not selected for the purposive synthesis remain visible and are distinguished from out-of-scope records and duplicate versions.

The final bibliography contains 88 distinct study/review records: 72 primary or foundational records and 16 reviews, of which 15 received full-text scope coding. Selection used a single review workflow and purposive relevance judgments, without independent dual screening. This is not an exhaustive systematic review or a meta-analysis; coverage limitations are consolidated in Section 11.

2.3. Extraction and Synthesis

For each retained record, the literature database stores title, authors, year, venue or repository, identifier, publication status, verification sources, method family, robot platform, task scope, data quantities, main findings, and limitations. Data quantities are accompanied by a unit and boundary: for example, successful episodes, human demonstrations, prior robot trajectories, or minutes of online interaction. Unavailable fields are marked as not reported or not extracted. Metadata follows the verified publication; technical extraction identifies the inspected version.

We compare mechanisms and conditions rather than rank headline success rates across papers. Quantitative values are retained only when their meaning can be located in a primary text. A reported comparison remains a within-study result unless the robot, task distribution, prior data, control interface, training budget, and evaluation protocol are demonstrably aligned. We distinguish a directly reported result from our interpretation and from a prospective recommendation. Reviews and emerging preprints receive explicit status labels; repository availability does not establish peer review.

3. The Real-World Data Bottleneck

3.1. A Demonstration Count Is an Incomplete Budget

Episodes differ in duration, action frequency, task difficulty, number of contacts, and operator involvement. Increasing the logging frequency increases the number of state–action pairs without necessarily adding independent behavioral coverage. Repeated near-identical demonstrations can teach execution consistency while adding little information about a new object or recovery state. Conversely, a short corrective segment can be highly valuable if it reaches a state that the policy encounters but a successful expert rarely visits. The sensitivity of manipulation learning to demonstration quality and model selection was already evident in the RoboMimic study [15].

Collection also involves setup, calibration, scene changes, failed attempts, labeling, inspection, and reset. DROID explicitly separates its successful trajectories from additional unsuccessful recordings [2]. BridgeData V2 combines teleoperated demonstrations with autonomously collected trajectories [1]. Episode counts require explicit definitions and separate labor records to serve as cost proxies.

3.2. Separate Accounting Boundaries

For a target deployment, we propose recording the resource vector

$$\mathbf{b} = (\mathbf{n}_{\text{prior}}, N_{\text{target}}, N_{\text{synthetic}}, H_{\text{human}}, H_{\text{robot}}, G_{\text{compute}}, N_{\text{test}}), \tag{1}$$

where $\mathbf{n}_{\text{prior}}$ is a source-stratified inventory in native units, such as robot episodes, human-video hours, and image–text pairs; N_{target} and $N_{\text{synthetic}}$ count newly acquired target and generated episodes, with physical or simulated provenance specified. Prior modalities are recorded separately, without conversion to equivalent robot episodes. H_{human} denotes person-hours, H_{robot} occupied physical robot-hours, and G_{compute} compute recorded separately by phase and hardware, including generation, training, and inference. N_{test} counts physical evaluation attempts. Person-hours and robot-hours can overlap in calendar time and must not be added as if they were disjoint durations. Human time should be subdivided into demonstration, reset, annotation, supervision, and integration. Episode counts require duration and sampling-rate context. Equation 1 is a reporting proposal, not an empirical law or a sufficient statistic for policy performance.

Two monetary boundaries are useful if defensible conversion rates are available:

$$C_{\text{marginal}} = C_{\text{acquire}} + C_{\text{adapt}} + C_{\text{integrate}} + C_{\text{validate}}, \tag{2}$$

$$C_{\text{allocated}} = C_{\text{marginal}} + \alpha C_{\text{upstream}}, \qquad 0 \le \alpha \le 1. \tag{3}$$

The first is relevant to a user downloading an existing model; the second attributes a declared share of pretraining and infrastructure costs. Both use a declared observation horizon. Recovery and maintenance within that horizon are assigned to the relevant component; each expense is counted once. An equal share $\alpha = 1/K$ is merely one accounting convention for K downstream uses, not an assumption that all uses benefit equally. Undocumented upstream expenditure remains unknown; allocation choices should distinguish shared investment from the marginal benefit of reuse. Figure 2 illustrates these boundaries.

ACCOUNTING BOUNDARIES

share alpha

	Upstream	→ Target acquire / adapt	→ Integration	→ Validate / recover
Data (native units)	Prior inventory	Target / synthetic	-	-
Person-hours	Collection / labels	Collect / supervise	Calibrate / deploy	Monitor / reset
Robot-hours	Upstream trials	Demos / online RL	Commissioning	Tests / recovery
Compute-hours*	Pretrain / simulate	Generate / tune	Deploy pipeline	Monitor / update
Physical test trials	Prior evaluations	Development tests	Acceptance setup	Held-out trials

Marginal user ledger: target + integration + validation / recovery, within each unit

Allocated lifecycle: marginal ledger + allocated upstream share alpha, within each unit

* Record accelerator / CPU type and count. Alpha is declared, not estimated here. Never sum across units.

Figure 2. Separate marginal-user and allocated-lifecycle ledgers. The marginal ledger records target acquisition/adaptation, integration, and validation/recovery. An explicitly declared share α of upstream investment is added only within the same unit for allocated accounting. Source-specific data quantities, person-hours, robot-hours, hardware-qualified compute-hours, and physical test trials remain separate; dashes indicate an unassigned cell, not zero cost.

3.3. The Unit of Benefit Must Also Be Specified

A policy that succeeds more often may execute more slowly, require more intervention, or cover a narrower task distribution. A suitable deployment objective therefore

constrains task success, time, intervention frequency, and failure severity, rather than optimizing demonstration count alone. Let $S = 1$ denote completion within a declared time limit without disallowed assistance. Then $\hat{p} = \sum_i S_i/n$ has an interpretable denominator only if aborted attempts, safety stops, and retries are assigned explicit outcomes. Partial task progress can be a useful additional measure, but it must not be silently relabeled as full-task success.

This distinction is particularly relevant to long-horizon systems. The evaluation of $\pi_{0.5}$ includes rubrics measuring task progress in previously unseen homes [34]. That provides evidence about useful behavior under new conditions, but a progress score is not automatically equivalent to an uninterrupted success probability. The value of a policy depends on which measure matches the application. For a repetitive machine-tending operation, completion without intervention and cycle time may dominate; for exploratory household assistance, partial progress and recoverability may also matter.

4. Demonstration Collection

4.1. Interfaces Change Both Cost and Data Distribution

Leader–follower teleoperation, handheld grippers, kinesthetic teaching, and human video expose different parts of an action. A robot-mediated interface records commands in a realizable action space and can expose the operator to robot limitations, but collection occupies the robot. A handheld interface moves acquisition into environments where transporting a robot is difficult, while introducing reconstruction and retargeting requirements. Human video can be abundant, but it generally omits robot actions, actuator limits, contact forces, and the counterfactual outcome of a different action. These sources offer distinct kinds of supervision rather than interchangeable episode counts.

ALOHA couples an accessible bimanual teleoperation system to Action Chunking with Transformers (ACT). The primary paper uses 50 demonstrations for most physical tasks and 100 for Thread Velcro, reporting about 10–20 minutes of recorded motion and 30–60 minutes of wall-clock collection per task [3]. This result applies to the evaluated interface, learner, and tasks. The reported collection interval includes resets and teleoperator mistakes; system construction, calibration, and other development effort remain separate.

UMI uses handheld grippers and a deliberately aligned observation–action interface to transfer human demonstrations to robots [5]. Its contribution includes relative action trajectories and inference-time latency matching, not portability alone. Robot-side sensing, inference, and execution delays can invalidate temporally aligned human-gripper data. Interface design therefore determines whether collected data can supervise deployable actions.

4.2. Distributed Collection and Diversity

BridgeData V2 broadens task and environment diversity on a shared low-cost platform, whereas DROID distributes a common robot setup across many scenes and institutions [1,2]. The latter's protocol encourages scene variation and records collection metadata, including success labels. Distribution reduces dependence on one laboratory's visual environment, while requiring consistent calibration, maintenance, transport, and operator training.

Mobile ALOHA extends demonstration collection to whole-body bimanual manipulation and studies co-training with static ALOHA data [35]. The benefit is not solely an increase in demonstrations of the target mobile task: the static dataset supplies reusable manipulation structure. Dobb-E, presented in *On Bringing Robots Home*, similarly connects portable acquisition, pretrained visual representations, and adaptation in homes [36]. To-

gether, these studies support analyzing collection and policy design jointly. Table 2 records representative quantities without treating them as a common benchmark.

Table 2. Acquisition interfaces and dataset boundaries. Counts describe inspected publications. Motion duration excludes setup and supervision. NR: no single comparable quantity reported or extracted; NR never means zero.

System	Interface / platform	Reported boundary	Residual obligation
ALOHA/ACT [3]	Bimanual leader–follower	50–100 demos/task; 10–20 min motion	Robot occupancy; setup/reset
GELLO [37]	Kinematically matched leader; several arms	User-study collection efficiency	Device construction; operator familiarity
UMI / DexCap [5,38]	Handheld gripper / hand mocap	Task-specific human motion; NR aggregate	Reconstruction; retargeting; latency
ARCap / DexUMI [39,40]	AR feedback / wearable hand interface	Physical policy evaluation; task-specific data	Feasibility feedback / hardware adaptation
BridgeData V2 [1]	WidowX; teleoperation + autonomous	50,365 human + 9,731 autonomous trajectories	Source mixture; platform coverage
DROID [2]	Distributed Franka teleoperation	76k successes; 350 recorded h; 564 scenes	Failures separate; transport/calibration
RoboMIND [41]	Four physical embodiments + simulation	About 107k successes include 30,035 simulated	Separate 5k physical failures; annotation
Mobile ALOHA / Dobb-E [35,36]	Mobile bimanual / home manipulation	Stage-specific prior and target data	Whole-body control; home integration

4.3. Coverage, Recovery, and Operator Expertise

Collection should be planned around the deployment distribution, not just a quota of successful trajectories. A policy may need variation in starting pose, distractors, object properties, camera placement, or the timing of contact. However, changing every factor in every episode can reduce the number of examples of a critical behavior and make failure diagnosis difficult. A practical design is to define a core task distribution and vary factors deliberately, preserving metadata that permits later stratification. The optimal allocation cannot be inferred from dataset size alone.

Recovery data deserve separate treatment. Smooth expert trajectories mainly occupy a successful subset of the state space. Deploying the learner can reveal states requiring correction; DAgger formalizes the importance of gathering supervision on the learner's visited distribution [14]. Physical implementation must additionally manage unsafe queries, delayed intervention, and irreversible failures. These requirements determine which learner-visited states can be queried on hardware. We recommend logging intervention timing and distinguishing a human rescue from an autonomous recovery when evaluating the resulting policy.

4.4. Collect Executable Information, Not Only More Motion

Collection interfaces should be compared per usable demonstration. GELLO's kinematically matched leader devices reduce the difficulty of commanding robot motion, while DexCap records human hand motion and retargets it to dexterous hardware [37,38]. They shift effort between operator control and reconstruction. ARCap makes that trade-off explicit: virtual-robot feedback warns users about kinematic or collision constraints during collection [39]. DexUMI instead designs a wearable mechanism around the target hand and adapts visual observations [40]. A robot-independent recording session can thus depend on substantial robot-specific design. The discriminating measurement is accepted, executable data per person-hour, including calibration, rejected recordings, and downstream correction.

Collection diversity also requires a design. Gao et al. show that policies can compose selected environmental factors, with prior robot data critical to their physical results [42]. Their analysis also identifies difficulty when factors interact physically. Independent variation of texture and object appearance can be cheaper than exhaustive combinations; combinations that change reachability or contact need direct coverage. This supports planned acquisition rather than uniformly increasing episode counts.

Multimodal and standardized corpora add complementary information. RH20T records force, audio, and visual observations; RoboMIND records explicit failure cases and applies a common collection protocol; AgiBot World expands coordinated collection and quality control [41,43,44]. These datasets expose different accounting boundaries. In the inspected RoboMIND text, the total of approximately 107,000 successful trajectories includes 30,035 simulation trajectories; the additional failure corpus is separate. Standardized formatting improves reuse, but does not make every trajectory an independent physical demonstration. Sensor synchronization, annotation, and failed attempts belong in the acquisition ledger.

Li et al. directly compare kinesthetic, VR, and spacemouse demonstrations on the same arm [45]. Kinesthetic teaching yielded strong downstream learning on drawer and glass tasks but weaker results on forceful sanitizer pressing, where replayed poses did not reproduce contact forces. Users also distinguished intuitive control from suitability for prolonged collection. This makes controller fidelity and operator burden necessary companions to demonstration-count comparisons.

5. Data Quality and Curation

5.1. Quality Is Conditional on the Learning Objective

Data quality includes observation fidelity, synchronization, action validity, task labeling, and coverage. It is not synonymous with visual neatness or shortest path length. The RoboMimic study compares human datasets of different quality and shows that algorithm choices and stopping criteria materially affect downstream performance [15]. A policy's validation loss can improve without a corresponding improvement in closed-loop success. Selecting checkpoints solely on an offline objective can therefore create an apparent data-efficiency gain that does not survive deployment.

Repeated actions may be redundant for behavior cloning but valuable for diagnosing contact timing. Failed trajectories can be inappropriate targets for unconditioned imitation while remaining useful for learning rewards, failure detectors, or offline-to-online value estimates. Removing all failures may improve the average imitation target and simultaneously remove information about recoverable states. Conversely, preserving every failure without indicating its outcome can teach undesirable behavior. The appropriate treatment depends on the learner and the intended function of the data.

5.2. Mixture Weighting and Trajectory Selection

ReMix addresses how to weight heterogeneous robot datasets using distributionally robust optimization, with normalization, discretization, and reference-model choices needed to make losses comparable across domains [46]. A domain can appear difficult because of an action-scale mismatch, observation ambiguity, or label noise, not because it offers the most transferable behavior. Weighting by an uncalibrated loss may then allocate more training effort to the wrong source.

Mutual-information-based curation estimates the usefulness of demonstrations through state diversity and action predictability [47]. It provides evidence that selecting data can improve learning, including on real ALOHA and Franka setups. Predictability and diversity depend on the chosen representation. Legitimate multimodal strategies can

appear unpredictable if the representation omits task context or history. Selection scores should therefore be checked against task context, alternative valid strategies, and recovery coverage. Complementary analysis by Parekh et al. shows why balancing imitation losses need not balance task success; its manipulation experiments use CALVIN simulation, so this supports a curation mechanism rather than a physical-deployment gain [48].

A quality score also needs an outcome interpretation. SIRIUS reweights human interventions and autonomous experience, while ThriftyDAgger requests assistance using novelty and risk estimates under a budget [49,50]. These mechanisms value visited failure states rather than merely expert-like trajectories. SIRIUS reports a limitation particularly relevant to cost accounting: reduced intervention fractions still required continuous human monitoring in its study. Human control time and human availability are therefore different resources. Discarding rare recoveries can improve imitation loss while reducing deployment-relevant information.

5.3. Preventing Leakage and Preserving Provenance

Episodes from the same scene, operator session, or source trajectory are statistically related. Random frame-level splitting can place near-identical observations in training and validation. Augmented descendants of a demonstration create a similar problem if their common ancestor crosses a split boundary. We recommend assigning the split at the level of the source episode, scene, or deployment factor before augmentation and recording parent identifiers for derived samples. Which grouping is appropriate depends on the generalization claim.

Data provenance should also include calibration versions, action conventions, collection frequency, source licenses, and filtering decisions. Cross-embodiment pooling particularly requires explicit normalization and representation choices [6,7]. A Cartesian displacement expressed in the camera frame is not interchangeable with a displacement in the robot base frame; changing a gripper convention can reverse the meaning of an action dimension. Such errors are integration failures even when the number of training examples is large. Curation is therefore part of system specification, not merely removal of low-quality images.

6. Augmentation and Synthetic Data

6.1. Observation-Space Augmentation

An observation transformation T_o that retains the original action label assumes that the desired action remains valid: $\pi^*(T_o(o)) = \pi^*(o)$ within the task and coordinate convention of interest. This is a local invariance assumption, not a general property of image editing. Changing background appearance may preserve an action; moving a target or altering a grasp affordance generally does not. Crops can remove a task-relevant object, and a visually plausible generated object can alter the geometry needed for contact.

GenAug instantiates semantically meaningful visual augmentation for tabletop manipulation using pretrained generative models and a constrained editing process [9]. Its primary text explicitly notes that observation augmentation cannot supply novel actions and that physical inconsistency can invalidate labels. The practical value is thus broader visual support around an existing behavior, provided the editing preserves functional validity. This makes it a plausible response to visual overfitting, not a replacement for demonstrations of new manipulation strategies.

RoboAgent combines semantic scene augmentation with a multi-task action-chunking policy and tests physical generalization [51]. Its intervention changes both the observations and the learner; the result therefore motivates joint design, while attributing the saving to augmentation alone requires the corresponding ablations.

6.2. Trajectory Transformation and Simulation

MimicGen decomposes demonstrations into object-centric segments, transforms them for new object configurations, stitches them, and executes the resulting trajectories to generate data [10]. It reports more than 50,000 generated demonstrations across 18 tasks from approximately 200 source demonstrations in its evaluated simulation and physical settings. This is a mixed-setting generated-data total, distinct from newly collected physical demonstrations. Generation depends on task decomposition, object information, executable motion, and successful completion checks. Its efficiency comes from reusing behavioral structure under those assumptions.

CyberDemo collects demonstrations in simulation, augments physical and visual conditions, and fine-tunes with a small amount of real data [11]. Its result challenges the assumption that directly collecting only in-domain real demonstrations is always the best use of operator time. The comparison is conditional on the availability of a usable simulator, the augmentation curriculum, and a transfer procedure. When these components already exist, simulation can be economical; constructing them for a new deformable material or unusual contact can change the cost substantially. A simulation-trained policy tested on hardware provides transfer evidence, but simulation interactions remain a separate budget category.

Trajectory reuse extends to bimanual coordination and recovery. DexMimicGen separates coordination dependencies between arms, whereas IntervenGen transforms corrective segments for errors such as inaccurate object pose estimates [52,53]. The latter's physical experiment transfers a simulation-trained block-grasping policy to a Franka robot; its scope is quasi-static rigid-object manipulation. DemoGen generates spatially transformed observations and actions from limited source demonstrations [54]. These methods expand particular behavioral neighborhoods, making source coverage and transformation validity more informative than generated count alone.

Simulation infrastructure supplies another reusable prior. Scaling Up and Distilling Down uses language-guided planning and privileged simulated state to generate labeled behavior; RoboCasa supplies scenes, assets, and trajectories for generalist learning; RoboTwin 2.0 combines automated generation with domain randomization [55–57]. Their physical transfer experiments make them relevant here, but they also reveal residual constraints. RoboCasa discusses successful generated trajectories with undesirable motion or collisions. In the inspected RoboTwin 2.0 physical table, adding synthetic data helps many conditions but hurts some individual task conditions. Acceptance by a generator and benefit to a physical learner are separate tests. Both generation quality and target mixing weights require validation.

6.3. New Embodiments, Self-Augmentation, and Dynamic Tasks

Recent work extends reuse beyond static visual changes. OXE-AugE augments robot embodiments in an existing cross-embodiment corpus and reports physical evaluations on new arm–gripper combinations [58]. SART uses demonstrator-annotated precision to guide self-augmentation from a limited source demonstration [59]. DynamicManip studies static-to-dynamic augmentation together with adaptive policy querying [60]. These approaches address expensive variations through different substitutions. Embodiment augmentation incurs retargeting and rendering assumptions; self-augmentation can consume physical interaction and validation; dynamic augmentation couples data diversity to timing and control design.

The 2026 AXIS preprint further illustrates how collection, filtering, augmentation, and evaluation can be packaged as a common data infrastructure [61]. Its quantitative evaluation is primarily simulation-based; qualitative Franka demonstrations in the appendix

do not establish quantified physical cost savings. It is therefore included as infrastructure context with limited physical evidence. The emerging status of these records and their boundaries is indicated in Table 3.

Table 3. Data expansion and validity checks. Physical evidence means hardware evaluation in the inspected text. P marks a repository preprint. Generated trajectories and physical demonstrations are separate quantities.

Method	Expanded support	Evidence boundary	Required checks
GenAug [9]	Visual observations	Physical tabletop; 10 seed demos in setup	Preserve action and affordance
MimicGen [10]	Object-relative segments	50k+ generated / about 200 seeds; mixed settings	Segmentation; stitching; success
DexMimicGen [52]	Bimanual coordination	Simulation generation; physical transfer	Coordination; digital twin; contact
DemoGen [54]	Spatial actions/observations	Physical policy evaluation	Geometric transforms; executable motion
IntervenGen [53]	Corrective trajectories	Simulation + physical block grasping	Pose error; rigid quasi-static scope
CyberDemo [11]	Visual/physical sim variation	Simulation demos + real fine-tuning	Simulator; curriculum; transfer
RoboCasa / RoboTwin 2.0 [56,57]	Scenes, assets, task trajectories	Simulation scale + selected physical tests	Trajectory quality; task-specific benefit
SART / OXE-AugE [58,59]	Precision / embodiment variation	Physical tests; OXE-AugE cited as P	Annotation / retargeting costs
DynamicManip / AXIS [60,61] (P)	Dynamics / data infrastructure	DynamicManip: physical; AXIS: qualitative hardware	Timing; no quantified AXIS physical savings

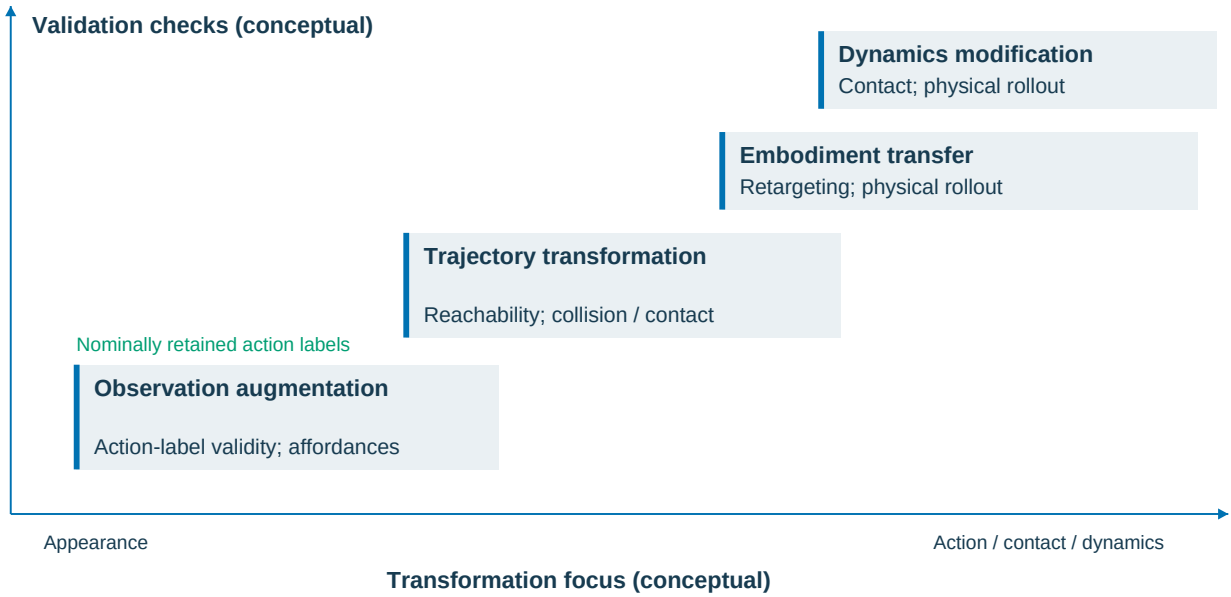


Figure 3. Conceptual augmentation-validity map. Region placement groups transformation types and their validation checks; neither axis represents a quantity, an ordering of costs, or a measured relationship. Observation edits require valid action labels; trajectory transformations require feasible motion and contact; embodiment and dynamics changes require retargeting and physical rollouts. A small visual edit can still invalidate a contact-critical affordance.

6.4. When More Synthetic Data Can Be Less Useful

Generated samples can share policy, simulator, or reconstruction errors; increasing quantity alone leaves missing contact modes, inadequate task specifications, and reward

errors unresolved. Synthetic data can amplify a narrow source strategy and overwhelm a small set of valid target examples. A useful experiment should therefore vary both synthetic quantity and mixing weight while keeping the target demonstration set fixed. It should compare against collecting additional real data at a matched resource budget, not only against using the initial dataset unchanged.

We recommend reporting the number of generation attempts, the fraction rejected, the acceptance criterion, and the physical validation burden. An acceptance rate is not equivalent to downstream usefulness: a generator may reliably produce easy trajectories that add little coverage. Conversely, low acceptance may be acceptable if it discovers rare but important recovery states cheaply in simulation. The appropriate synthetic-to-real ratio therefore depends on coverage, validity and validation cost.

IGen further generates paired observations and planned actions from an image and fine-tunes a pretrained π_0 policy for physical Franka tasks [62]. Its reported comparison under matched collection time is relevant to marginal acquisition efficiency. The pretrained policy, geometric reconstruction, motion planning, and generation compute remain outside a demonstration-count saving; this is not a matched-lifecycle-cost result.

7. Data-Efficient Policy Learning and Adaptation

7.1. Task-Specific Imitation and Structured Actions

Behavior cloning (BC) minimizes a supervised discrepancy between demonstrated and predicted actions. Its simplicity avoids explicit reward construction, but the training distribution need not cover states encountered after the policy's own errors. DAgger addresses this mismatch through interactive supervision [14]; expressive policy classes and better representations address different aspects of the problem. No action model removes the need to specify what should happen in an unseen failure state.

ACT predicts action chunks and uses temporal aggregation in a bimanual imitation system [3]. Diffusion Policy models a conditional action distribution through denoising and combines it with receding-horizon execution [4]. Both exploit temporal structure, but their capacity to represent action sequences is distinct from physical generalization. Increasing chunk length can reduce repeated prediction overhead while delaying feedback. The relevant choice depends on task dynamics, the duration of contact transitions, and observation latency, not only offline action error.

Geometry can reduce the burden of learning invariances. PerAct predicts spatial actions in a voxelized representation and relies on a motion planner for execution [63]. RVT aggregates information through rendered views of 3D observations [64]. DP3 uses a compact point-cloud representation with diffusion-based action generation [65]. Their data requirements must be interpreted alongside depth sensing, camera calibration, workspace assumptions, and control interfaces. Predicting a key pose with a motion planner is a different learning problem from generating dense motor commands, even if both are called manipulation policies.

The action interface can itself explain apparent efficiency. HYDRA combines sparse waypoints with dense actions, assigning different execution structures to different task phases [66]. RISE predicts continuous actions from point clouds and reports sensitivity of comparisons to demonstration style and geometric encoder choices [67]. Thus, a result attributed to "3D versus 2D" may partly reflect control abstraction, demonstration conventions, or model capacity. A useful ablation holds these fixed while changing the observation representation.

Execution efficiency is an additional axis. Consistency Policy distills a diffusion teacher; VQ-BeT compresses actions into a learned discrete representation [68,69]. Both offer alternatives to iterative action generation, with physical evaluations. Distillation

spends teacher-training and student-training compute to reduce inference burden. It improves data efficiency only if a matched-data experiment also shows a better physical learning curve; faster action generation by itself answers a different question.

7.2. Pretrained Representations and Foundation Policies

Visual representation pretraining can reduce the need to learn perception from target demonstrations. R3M transfers representations learned from human video to manipulation, including a real-apartment evaluation with limited demonstrations [70]. The artificial-visual-cortex study evaluates pretrained visual representations across embodied tasks and emphasizes the dependence of results on adaptation choices [71]. These results motivate reuse, but a perception prior does not provide every element of control: action mappings and task-specific feedback still require supervision or interaction.

RT-1 and RT-2 illustrate different routes to scaling robot policies: large-scale robot demonstration corpora and transfer from vision–language models into action prediction [72,73]. Open X-Embodiment broadens the sources of robot experience [6]. Octo supports adaptation to new observation and action specifications, while OpenVLA fine-tunes a pretrained multimodal backbone for robot control [7,8]. Their comparatively small target datasets should be reported together with their prior robot experience. Downloadable weights make this reuse valuable to a downstream laboratory even when total upstream data are large.

The distinction between parameter efficiency and data efficiency is essential. Updating a low-rank subset of weights can reduce adaptation memory and compute without necessarily reducing the required target demonstrations. OpenVLA studies such parameter-efficient adaptation, while the optimized fine-tuning (OFT) study changes action decoding, chunking, representation, and objectives to improve both performance and speed [8,74]. The latter comparison evaluates a joint training and execution recipe; isolating target-data savings requires matched-data learning curves.

Flow-based and hybrid systems such as π_0, $\pi_{0.5}$, and GR00T N1 further combine diverse robot data, pretrained knowledge, and expressive action prediction [34,75,76]. They broaden the behavioral repertoire, but their integration does not make semantic familiarity equivalent to contact competence. Table 4 compares the principal forms of reuse and their residual obligations.

FAST illustrates why action tokenization belongs in the resource ledger: compressing action sequences changes training cost and the feasibility of learning high-frequency behavior [77]. Its universal tokenizer also has a distinct prior-data investment. SmolVLA combines a compact VLA with an asynchronous execution stack and community data [80]. Small models can reduce access barriers, but data provenance, robot compatibility, and end-to-end latency remain relevant. Comparisons should separate backbone size, action representation, prior corpus, and execution scheduler instead of treating all gains as fewer demonstrations.

7.3. Few-Shot Adaptation Means Different Things

Few-shot learning may denote fine-tuning on a small target dataset, retrieving a relevant behavior, conditioning on a demonstration, or learning a residual policy after a few examples. It should always specify which parameters change, whether test-time interaction is permitted, and what prior tasks were seen. FISH begins with a weak policy derived from a small number of demonstrations and learns a residual through subsequent online interaction [12]. MimicPlay uses human play to inform long-horizon behavior while retaining robot supervision for execution [81]. These methods economize different kinds of supervision.

Table 4. Learning mechanisms, reused information, and remaining costs. Parameter efficiency, target-data efficiency, and execution speed are separate outcomes.

Family / examples	Reused information	Target requirement	Trade-off
ACT / Diffusion / HYDRA [3,4,66]	Temporal and action structure	Representative demonstrations	Action abstraction versus feedback
PerAct / RVT / DP3 / RISE [63–65,67]	Spatial structure	Depth, calibration, demonstrations	Geometry versus sensing burden
R3M / VC-1 [70,71]	Pretrained visual features	Motor supervision	Perception transfer versus control gap
RT-1 / Octo [7,72]	Prior robot experience	Compatible interface; adaptation	Marginal savings; upstream investment
RT-2 / OpenVLA / OFT [8,73,74]	Visual, semantic, and robot priors	Fine-tuning and execution integration	Alignment; tuning; latency
π_0 / $\pi_{0.5}$ / GR00T N1 [34,75,76]	Heterogeneous priors; action heads	Physical task validation	Broad repertoire versus reliability
Consistency / VQ-BeT / FAST [68,69,77]	Distillation / action compression	Teacher or tokenizer training	Compute savings; data savings assessed separately
FISH / HIL-SERL / ResFiT [12,13,78]	Initial policy and online feedback	Robot trials; rewards; resets	Fewer demos; more interaction
DPPO [79]	Demonstration-initialized diffusion	Simulation RL; physical transfer	Simulator throughput; transfer validity

Narrow behaviors, stable interfaces, and adequate demonstrations favor task-specific policies; aligned observation, action, and task priors favor generalist initialization. Model size alone is insufficient: a poorly aligned large model incurs adaptation engineering, while a compact model may exploit a suitable spatial prior.

7.4. Reinforcement Learning from Demonstrations and Corrections

Demonstrations can initialize an RL policy, guide exploration, populate replay, define an imitation-derived reward, or provide corrections during training. DAPG establishes a demonstration-augmented policy-gradient approach to dexterous manipulation [16]. RLPD studies simple but consequential choices for combining offline and online experience, including symmetric sampling and critic regularization [82]. Its benchmark evidence should not itself be counted as a direct physical deployment study; its relevance includes its use within later real-robot systems.

HIL-SERL combines prior demonstrations, human corrections, efficient off-policy learning, a pretrained visual backbone, and low-level control design for dexterous physical tasks [13]. Its reported learning times show that real-world RL can be practical under a carefully integrated system. The comparison with BC matches human-provided data, but RL adds autonomous physical experience. Its improvement beyond imitation must therefore be assessed alongside the additional interaction and operator involvement.

DPPO demonstrates a different route: reinforcement learning fine-tunes a diffusion policy, including simulation training followed by physical transfer [79]. Its exploration and optimization benefits must be evaluated against hardware interaction costs when selecting an on-policy method for a physical system. The availability of parallel simulated environments changes the cost of samples. Matching demonstration counts while ignoring whether subsequent interaction occurred in simulation or reality hides this distinction.

7.5. Recovery Supervision and the Cost of Autonomy

Recovery can be collected before or during deployment. DART perturbs expert execution to elicit corrective behavior; ThriftyDAgger gates requests during policy execution; SIRIUS learns from the resulting mixture of autonomous and human-controlled segments [49,50,83]. These methods address distribution shift through different human interfaces. Their comparison should include monitoring time, request frequency, intervention duration, and post-intervention autonomous success.

RoboFuME pretrains policies and reward inference to reduce the need for task-specific supervision during physical fine-tuning [84]. Its experimental setup still includes periodic manual resets and interruptions, so "reset-free" describes an algorithmic direction rather than zero human maintenance. ResFiT applies off-policy residual learning around a behavior-cloned base policy, including physical bimanual tasks [78]. It offers a route to preserve useful behavior while improving errors, with additional interaction, reward, and hardware safeguards. Together with HIL-SERL, these studies make autonomous learning a system property: reward errors, resets, and recovery can dominate the marginal cost even when the demonstration count is small.

8. Generalization and Real-World Deployment

8.1. State What Was Withheld

Generalization can concern new object instances, appearances, geometries, instructions, task compositions, environments, embodiments, or contact dynamics. STAR-Gen shows the value of separating visual, semantic, and behavioral generalization and provides real-world case studies [32]. A new background does not establish a new manipulation strategy; a new task name does not establish unseen contact dynamics; a new robot may still share the same end-effector interface and reachable workspace. Table 5 maps useful claims to the corresponding holdouts and evidence limits.

Cross-task claims also depend on how tasks are defined. Recombining known actions with different language can test semantic composition, while learning a new tool use may demand a new action distribution. Cross-environment evaluation is stronger when scenes are excluded from collection and tuning, not merely photographed under a new lighting condition. Cross-embodiment evaluation must specify whether adaptation data, camera recalibration, or a new action head were supplied. "Zero-shot" is meaningful only relative to the excluded resource: zero target demonstrations does not imply zero target engineering.

8.2. Simulation Is Useful but Conditional Evidence

LIBERO supplies a simulation framework for studying knowledge transfer and lifelong robot learning [85]. The Colosseum systematically perturbs conditions and includes paired physical experiments, while SIMPLER investigates how simulated evaluations can track behavior on particular real robot setups [86,87]. These tools improve reproducibility and allow inexpensive stress tests. Their sim–real correlations are empirical findings under the tested systems and tasks, not guarantees for a new embodiment, deformable object, or contact regime.

The 2026 benchmark audit identifies shortcut solvability, inadequate statistical discrimination, overfitting, and dependence on data sources as threats to interpreting benchmark scores [88]. This preprint reinforces a testable requirement: an evaluation must discriminate the capability claimed. A language-conditioned policy should be tested on tasks where language changes the correct action, not only on scenes whose layout already reveals the goal.

Table 5. Generalization claims and required physical holdouts, building on STAR-Gen, the Colosseum, and SIMPLER. Separate the factor withheld from the outcome measured.

Claim	Hold out	Evidence examples	Measure separately
Visual robustness	Lighting, background, distractors	GenAug; Colosseum [9,86]	Contact strategy
Object transfer	Instances and geometry factors	UMI; DP3 [5,65]	Affordance/dynamics changes
Semantic transfer	Instructions changing correct action	RT-2; STAR-Gen [32,73]	Goal selection; execution
Environment transfer	Whole scenes/homes from tuning	DROID; $\pi_{0.5}$ [2,34]	Progress; autonomous completion
Embodiment transfer	Platform; disclose adaptation	OXE; Octo; OXE-AugE [6,7,58]	Calibration; controller changes
Composition	New combinations or sequences	Gao et al.; MimicPlay [42,81]	Factor interactions; recovery
Sustained use	Later sessions; drift/fault conditions	Proposed extension of physical tests	Interventions; cycle time; maintenance



Figure 4. Evidence envelopes and the questions they support. Each envelope has explicitly stated conditions rather than an ordinal score. Simulation supports reproducible screening, matched hardware tests feasibility, held-out physical conditions transfer, and repeated operation persistence. Broader coverage alone provides no safety certification.

8.3. Latency, Contact, and Recovery

Successful offline prediction is compatible with failed closed-loop control if observations are stale or action chunks outlast a critical event. UMI explicitly addresses latency matching, and OFT changes the action-generation pathway to enable higher-frequency operation [5,74]. End-to-end timing should include capture, preprocessing, transfer, inference, command delivery, and execution. Reporting action throughput alone can hide gaps between chunks or a long sensing-to-action delay.

Contact-rich manipulation adds observability and compliance issues. Vision may not reveal the force needed to distinguish a jammed insertion from a successful one, and an accurate position command can still produce an unsuitable contact force. The force-oriented survey documents approaches that incorporate tactile or force information and relates them to low-level control [24]. Additional sensing can reduce ambiguity while creating calibration and transfer costs. We therefore recommend evaluating the marginal benefit of additional sensing under matched data, rather than assuming either that vision is sufficient for every contact task or that tactile sensing is automatically data-efficient.

Recovery is a deployment performance measure. Repeated deployment should record autonomous retries, human resets, aborted episodes, damaged objects, and soft-

ware or calibration changes. Policy errors require adaptation; perception, controller, or timestamp errors require corresponding system corrections. Figure 5 expresses a diagnosis-led collection loop.

9. Comparative Analysis

9.1. What Can Be Compared Quantitatively

Table 6 collects selected quantities whose boundaries can be recovered from primary texts. Budgets of 40 real demonstrations per task in DP3, 50–100 in ACT, and approximately 100 for Octo adaptation do not establish which method is most efficient, because the robots, tasks, prior experience, observation spaces, and success criteria differ [3,7,65]. A fair ranking would need learning curves at multiple target budgets under a common protocol.

Some within-study comparisons are more informative. ReMix changes data mixture selection; OFT studies the action and optimization recipe; HIL-SERL compares imitation and interactive refinement; CyberDemo contrasts ways of sourcing and expanding demonstrations [11,13,46,74]. Their conclusions depend on the components varied and the evaluation setting. For example, OFT changes several interacting components, and the simulation headline cannot be relabeled as a physical success rate. A controlled study can establish the utility of a recipe without identifying every causal contribution independently.

Table 6. Selected primary-text quantities. Rows have different tasks, priors, and controllers, and therefore support within-study interpretation, not cross-paper ranking. All quantities are reported rather than rerun.

Study	Quantity / boundary	Interpretation	Unmeasured or extra cost
ACT [3]	50–100 demos/task; 10–20 min motion	Limited-data bimanual imitation	30–60 min collection; setup/development
DP3 [65]	40 demos/physical task; four tasks	Compact geometry with diffusion	Depth/calibration; 10-demo result is simulation
Octo [7]	800k prior episodes; around 100 target	Reuse at small target budgets	Upstream collection; target integration
OpenVLA [8]	970k prior robot trajectories	Robot and multimodal pretraining	Web/video priors; fine-tuning
FISH [12]	1–3 demos; up to 20 min interaction	Few demos initialize refinement	Online robot use
HIL-SERL [13]	1–2.5 h reported physical training	Corrections plus RL improve behavior	BC comparator lacks autonomous RL trials
OpenVLA-OFT [74]	LIBERO: 76.5% to 97.1%; 26× action throughput	Recipe improves reported comparison	Simulation success; throughput excludes full latency

9.2. Seven Conditional Resource Comparisons

More data versus better data. Broader acquisition is favored when a required object, state, or recovery behavior is missing. Selection is favored when available data are redundant, inconsistent, or imbalanced. OXE and OpenVLA demonstrate broad reuse; ReMix and DemInf show the value of weighting or selecting that experience [6,8,46,47]. The hidden cost of selection is scoring and validation, and aggressive filtering can remove rare behavior. A discriminating experiment crosses corpus size with selection strategy at fixed training compute, then evaluates missing-domain transfer and recovery separately. Selection addresses the information already present; missing capabilities still require an additional source of supervision or interaction.

Task-specific versus pretrained policies. Task-specific learning is favored for a stable, narrow skill with compatible demonstrations and a tractable action space. Pretraining is favored when its task, observation, and action priors align with several target skills.

ACT and DP3 support the former regime; Octo and OpenVLA support the latter [3,7,8,65]. The hidden variables are upstream data, adaptation search, and interface engineering. Compare both initializations at several identical target budgets, with a declared tuning allowance and common success criteria. Report performance against target labor and robot occupancy as well as demonstrations. This would establish where reusable priors repay their integration cost.

2D visual priors versus 3D geometric priors. Rich visual pretraining is favored when recognition and semantic composition dominate. Explicit geometry is favored when spatial relations must be learned from limited target examples and depth is reliable. R3M and foundation policies transfer visual knowledge; PerAct, DP3, and RISE exploit different spatial and execution structures [63,65,67,70]. Depth calibration, missing surfaces, and controller choice are hidden costs. Hold the action interface, demonstrations, and tuning budget fixed, vary the encoder and sensing inputs, and test appearance changes separately from geometric and contact changes. RISE's discussion of demonstration conventions warns that unmatched action representations can obscure the mechanism behind a comparison.

Offline imitation versus interactive refinement. Offline learning is favored when robot access is scarce, resets are expensive, or exploration is difficult to supervise. Interactive refinement is favored when recurring errors lie outside expert support and additional trials are affordable. FISH, HIL-SERL and ResFiT show how demonstrations can seed further improvement; RoboFuME exposes the importance of reward and reset infrastructure [12,13,78,84]. Autonomous interaction and monitoring remain costs. Compare offline acquisition, corrective imitation, and RL at matched person-hours, and separately at matched robot-hours. A method can win one comparison and lose the other; the difference is operationally useful rather than contradictory.

Synthetic scale versus physical validity. Synthesis is favored when source behavior is adequate and the missing variation can be transformed reliably. New physical collection is favored when the deficit is an unknown contact mode, dynamics, or infeasible retargeting. GenAug, MimicGen, DemoGen and IntervenGen supply different neighborhoods of valid variation [9,10,53,54]. Generator engineering and rejection tests are hidden investments. Hold seed data fixed, vary generation scale and validity filtering independently, and compare the result with additional real collection at matched labor. Evaluate both the targeted shift and adverse effects on nominal behavior; RoboTwin's condition-dependent physical results motivate this second check [57].

Semantic generalization versus physical competence. Semantic priors are favored when the robot recognizes the scene but selects the wrong goal or subtask. Additional geometric, contact, or corrective information is favored when it selects the correct goal but cannot execute it. RT-2, $\pi_{0.5}$ and STAR-Gen motivate separating these capabilities [32,34,73]. The hidden cost is grounding and verifying intermediate decisions. Vary instructions alongside physical perturbations while preserving each factor's causal role in the task. Score goal selection, execution, and full completion separately. This design distinguishes better reasoning from improved motor competence and identifies when both are required.

Demonstration efficiency versus total system efficiency. Minimal new supervision is favored when demonstrations are the dominant constraint and reusable infrastructure already exists. A larger target dataset can be preferable when it avoids costly simulation construction, unstable tuning, or frequent recovery. FAST and Consistency Policy illustrate computation substitutions, while UMI and ARCap illustrate collection/integration substitutions [5,39,68,77]. The required evidence is a lifecycle learning curve with separate units and declared upstream allocation. Run a repeated-use study across target tasks,

logging development, physical tests, recovery, and maintenance. The key question is the reuse horizon at which initial investments become worthwhile.

9.3. A Practical Reporting and Decision Protocol

We propose beginning with a named deployment envelope: robot, end effector, sensing and control interface, object range, environment variation, time limit, allowed assistance, and unacceptable outcomes. A target demonstration budget should be accompanied by occupied robot time, human time, and prior sources. Model selection uses a validation set disjoint from the final held-out conditions. Data descendants remain grouped with their source episode, and test failures are not added to training until the evaluation round closes.

Next, compare a task-specific baseline with a compatible pretrained initialization using the same target data and an explicit tuning budget. Add augmentation or interaction only when failure analysis suggests which missing information it can provide. Report the result at several target budgets when feasible, and record variation across training seeds separately from variation across physical scenes or sessions. Table 7 turns this proposal into a compact comparison contract. Application-specific acceptance requirements remain necessary.

Table 7. Proposed minimum comparison contract for data-efficient physical manipulation. This contract records the resources needed to interpret a claim.

Item	Record before comparison	Purpose
Deployment envelope	Robot, sensors, controller, tasks, assistance, time limit	Fix capability and acceptance criteria
Prior resources	Corpora/units, model version, overlap, simulator/assets	Identify transferred investment
Acquisition	Episodes/duration; person-hours; robot-hours; failures	Expose labor and occupancy
Derived data	Parents, transformation, attempts, acceptance checks	Track validity and split leakage
Adaptation	Trainable weights, hardware, tuning, online trials	Separate data, compute, and interaction
Evaluation	Holdout factors, scenes/sessions, seeds, all attempts	Identify independent evidence units
Execution/recovery	Latency, cycle time, interventions, retries, aborts	Measure practical physical benefit
Decision	Acceptance criteria; added burden; residual failures	Accept or re-diagnose; freeze before final test

Figure 6 maps the reviewed mechanisms to resource demands.

10. Open Challenges and Future Directions

10.1. Measure the Full Learning Curve in Resources

Demonstration-only learning curves leave human time, robot occupancy, and computation unresolved. Paired curves in these resources would expose the corresponding trade-offs. A small number of budget points can reveal whether an approach mainly improves the low-data regime, raises the performance ceiling, or pays a large fixed cost before becoming attractive. A factorial experiment should cross several target-data budgets with pretrained versus task-specific initialization, maintain a common physical evaluation distribution, and report marginal and allocated upstream costs when known. Missing cost components should be explicit. This would turn a qualitative claim of data efficiency into a decision-relevant comparison without requiring a universal currency conversion.

The unit of analysis also matters. Repeated trials on one trained checkpoint do not quantify training instability. Repeated starts in one scene do not quantify cross-home generalization. A useful design samples checkpoints, scenes, and sessions separately and

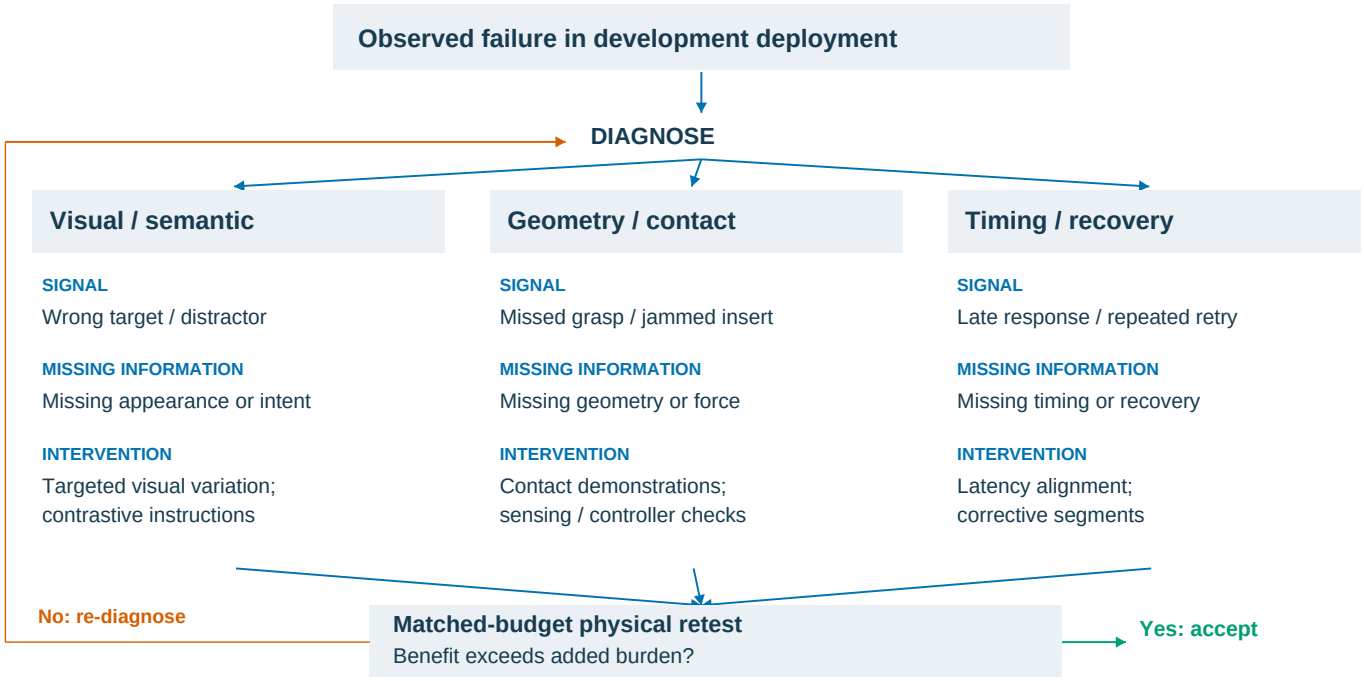


Figure 5. Diagnosis-led intervention during development and deployment. Signals suggest missing information and candidate changes; matched-budget physical retesting checks whether benefit justifies the added burden. Rejected changes return to diagnosis. Development-validation trials support this loop; final claims require untouched physical test conditions after the recipe is frozen.

RESOURCE SUBSTITUTION BY MECHANISM

	Acquire	Curate	Augment	Learn / adapt	Interactive refine
Target demos	Faster capture	Select / reweight	Reuse sources	Structure / prior P	Correct errors I
Human effort	Interface design	Review / labels	Seed / validate S	Tuning / align	Supervise / reset
Robot interaction	Collect I	Reuse logs	Sim or replay S	Prior + target P	Online trials I
Compute	Reconstruct	Score / train	Generate S	Train / tune P	Update policy
Deployment burden	Calibration	Provenance	Validity tests	Control / latency	Recovery R
Examples	ALOHA, UMI, DROID	ReMix, DemInf	GenAug, MimicGen, CyberDemo	ACT, Diffusion, DP3; RT-1/2, Octo, OpenVLA, pi0	HIL-SERL, FISH; DPPO (sim RL)

P: transferred prior S: synthetic assumption I: physical interaction R: recovery obligation

Figure 6. Taxonomy of resource substitutions. Cells identify mechanisms or expenditures rather than score methods; the same method can occupy multiple stages. The learning/adaptation column includes task-specific action structure as well as transferred pretraining. Examples have physical evaluations in the cited texts, while DPPO's refinement uses simulation before physical transfer. Tags highlight priors, synthetic assumptions, physical interaction, and recovery obligations; actual evidence boundaries remain in Tables 3–6.

states which source of variation an interval represents. When resources are limited, the study should narrow the generalization claim rather than count correlated trials as broad deployment evidence.

10.2. Validate Data Selection Under Rare Failures

Quality estimators can undervalue rare states, recovery behavior, and legitimate multimodality; these cases therefore require separate evaluation. A controlled experiment can inject mislabeled actions, redundant successes, and valid recovery segments separately into a fixed corpus, then measure both nominal success and recovery. This would distinguish noise filtering from removal of difficult examples. The same experiment should evaluate whether scores remain useful when the policy architecture or observation representation changes. ReMix and mutual-information curation provide concrete starting points, but their success does not eliminate this transfer question [46,47].

10.3. Attach Validity Evidence to Synthetic Data

Generated datasets should carry parent-episode identifiers, transformation parameters, rejection criteria, and the specific physical assumptions that remain untested. Research should distinguish visual plausibility, kinematic feasibility, dynamic plausibility, and task success rather than collapse them into one quality score. A useful stress test deliberately alters an affordance or contact parameter while preserving appearance, then checks whether the augmentation pipeline detects label invalidity. For a novel embodiment, comparisons should isolate the benefit of rendering diversity from the benefit of correct action retargeting.

Learned world models could support generation, planning, and evaluation, as the contemporary survey documents [33]. Their reuse creates a particular dependence risk: training and evaluating in the same learned model may reward its own errors. A held-out physical validation set is therefore essential to establish whether an apparent gain transfers beyond the model. Independent physical tests can expose errors shared by generation and evaluation.

10.4. Learn When to Acquire a New Demonstration

Fixed collection quotas ignore the changing value of additional data. Active collection could prioritize uncertainty, recovery states, new contacts, or underrepresented task conditions. However, predictive uncertainty can be poorly calibrated under distribution shift, and an uncertain state may simply be unobservable with the current sensors. A practical active-learning policy should therefore include abstention and request a different sensing or control intervention when appropriate. Randomize uncertainty-based, failure-based, and simple diversity-based collection across identical initial policies and person-hour budgets. Measure accepted episodes, time spent waiting or supervising, autonomous held-out success, and recovery. This tests whether sophisticated query selection saves usable human effort rather than merely issuing fewer requests.

Demonstration and language interfaces may also work together. A person can identify an incorrect subtask through language while providing a short physical correction for execution. Such mixed supervision requires separate accounting: a semantic correction and a full trajectory are not the same label. Long-horizon systems such as $\pi_{0.5}$ motivate studying how high-level instruction and low-level supervision complement one another [34]. The open question is which intervention most efficiently resolves a diagnosed error, not whether a single feedback modality can replace all others.

10.5. Generalize Across Contact and Embodiment Without Hiding Engineering

Cross-embodiment learning will require explicit descriptions of action semantics, actuator constraints, compliance, sensing, and latency. It is insufficient to normalize action dimensions and assume that two platforms share the same physical capabilities. Benchmarking should report both adaptation data and engineering changes, including new controllers or motion planners. Separating shared policy knowledge from platform-specific interfaces would make negative transfer easier to diagnose and avoid overstating hardware independence.

Contact-rich tasks are a particularly valuable test because they can expose gaps hidden by appearance-based evaluation. Matched studies should compare vision-only sensing with additional force or tactile sensing while reporting calibration and collection effort. They should evaluate changes in object stiffness or friction separately from changes in appearance. Whether richer sensing saves demonstrations enough to offset its integration burden is an empirical question, with likely task-dependent answers [24].

10.6. Evaluate Sustained Physical Use

Most controlled experiments answer whether a learned behavior works within an evaluated setting. Sustained use additionally asks how often the system needs intervention, how it responds to drift, and how recovery affects throughput. Evaluate frozen policy recipes across repeated operating sessions with prespecified drift and fault conditions. Record time to intervention, causes of aborts, autonomous recovery success, maintenance or recalibration events, and completed tasks per occupied robot-hour alongside ordinary task success. A policy that completes a difficult sequence once may still be unsuitable for repetitive unsupervised operation.

The reporting protocol should not obscure failure severity by averaging across tasks. A benign missed grasp and a damaging contact have different implications. Deployment efficiency must be conditioned on application-defined acceptance criteria and tested under those criteria. Data-efficient learning becomes useful when reduced acquisition effort produces a behavior that remains acceptable after the cost of verification and recovery is included.

11. Limitations of This Review

The search uses bounded public-source retrieval and a purposive candidate register rather than exhaustive database retrieval. It can underrepresent inaccessible, negative, or industrial deployment evidence. Screening and synthesis used one workflow, without independent dual screening or a preregistered protocol. Rapidly changing preprints and differences between repository and final versions constrain publication-status and technical comparisons; the extraction register identifies the inspected sources. Heterogeneous robots, tasks, priors, controllers, and evaluation units prevent a meaningful pooled effect estimate. Primary studies also omit labor, integration, and maintenance variables needed for complete lifecycle accounting. The proposed ledger and reporting protocol are analytical recommendations whose practical utility requires prospective evaluation. The audit trail supports inspection within these coverage and attribution limits.

12. Conclusions

Limited real robot data can be used more effectively through better collection interfaces, deliberate curation, valid augmentation, structured policy learning, pretrained knowledge, and interactive refinement. These mechanisms substitute different resources and rely on different assumptions. Demonstration counts alone cannot determine which is most economical or most likely to generalize.

The resource-substitution taxonomy links these mechanisms to distinct information deficits. Its contribution relative to existing reviews is an integrated decision structure connecting a claimed saving to the expenditure boundary and the physical evidence still required. Separate marginal-user and allocated-lifecycle ledgers distinguish downstream costs from shared upstream investment, while the seven conditional comparisons connect those boundaries to deployment decisions. The appropriate policy family then depends on the diagnosed deployment deficit and the available resource budget. Visual augmentation addresses a different deficit from contact demonstrations; semantic pretraining supplies different knowledge from a calibrated controller; a benchmark result supports a different claim from sustained physical operation. The deployment-oriented validation protocol calls for matched-budget, clearly delimited physical evidence that records interventions and recovery, with uncertainty and missing cost information made explicit.

Supplementary Materials: Supplementary File S1: Literature database; Supplementary File S2: Search strategy and query audit; Supplementary File S3: Final screening register, citation tracing and source-verification metadata; Supplementary File S4: Figure-generation and extraction source code. They contain literature metadata and analytical extraction, not newly collected robot data.

Author Contributions: The author confirms sole responsibility for the conceptualization, methodology, literature investigation, analysis, manuscript preparation, revision, and final approval of the submitted manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable. This article reviews published literature and reports no new human-participant or animal study.

Informed Consent Statement: Not applicable.

Data Availability Statement: The review's literature database, search documentation, and extraction notes are supplied with the submission package. No new experimental datasets were generated. The cited original datasets remain available subject to their respective providers' access conditions and licenses.

Conflicts of Interest: The author declares no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

BC	Behavior cloning
IL	Imitation learning
RL	Reinforcement learning
VLA	Vision-language-action
VLM	Vision-language model
OXE	Open X-Embodiment
OFT	Optimized fine-tuning

References

1. Walke, H.R.; Black, K.; Zhao, T.Z.; Vuong, Q.; Zheng, C.; Hansen-Estruch, P.; He, A.W.; Myers, V.; Kim, M.J.; Du, M.; et al. BridgeData V2: A Dataset for Robot Learning at Scale. In Proceedings of the Conference on Robot Learning, 2023, Vol. 229, pp. 1723–1736, [2308.12952].
2. Khazatsky, A.; Pertsch, K.; Nair, S.; Balakrishna, A.; Dasari, S.; Karamcheti, S.; Nasiriany, S.; Srirama, M.K.; Chen, L.Y.; Ellis, K.; et al. DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset. In Proceedings of the Robotics: Science and Systems XX, 2024, Vol. 20, [2403.12945]. https://doi.org/10.15607/RSS.2024.XX.120.

3. Zhao, T.Z.; Kumar, V.; Levine, S.; Finn, C. Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware. In Proceedings of the Robotics: Science and Systems XIX, 2023, Vol. 19, [2304.13705]. https://doi.org/10.15607/RSS.2023.XIX.016.
4. Chi, C.; Xu, Z.; Feng, S.; Cousineau, E.; Du, Y.; Burchfiel, B.; Tedrake, R.; Song, S. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research* **2025**, *44*, 1684–1704, [2303.04137]. https://doi.org/10.1177/02783649241273668.
5. Chi, C.; Xu, Z.; Pan, C.; Cousineau, E.; Burchfiel, B.; Feng, S.; Tedrake, R.; Song, S. Universal Manipulation Interface: In-The-Wild Robot Teaching Without In-The-Wild Robots. In Proceedings of the Robotics: Science and Systems XX, 2024, Vol. 20, [2402.10329]. https://doi.org/10.15607/RSS.2024.XX.045.
6. O'Neill, A.; Rehman, A.; Maddukuri, A.; Gupta, A.; Padalkar, A.; Lee, A.; Pooley, A.; Gupta, A.; Mandlekar, A.; Jain, A.; et al. Open X-Embodiment: Robotic Learning Datasets and RT-X Models. In Proceedings of the 2024 IEEE International Conference on Robotics and Automation (ICRA), 2024, pp. 6892–6903, [2310.08864]. https://doi.org/10.1109/icra57147.2024.10611477.
7. Ghosh, D.; Walke, H.R.; Pertsch, K.; Black, K.; Mees, O.; Dasari, S.; Hejna, J.; Kreiman, T.; Xu, C.; Luo, J.; et al. Octo: An Open-Source Generalist Robot Policy. In Proceedings of the Robotics: Science and Systems XX, 2024, Vol. 20, [2405.12213]. https://doi.org/10.15607/RSS.2024.XX.090.
8. Kim, M.J.; Pertsch, K.; Karamcheti, S.; Xiao, T.; Balakrishna, A.; Nair, S.; Rafailov, R.; Foster, E.P.; Sanketi, P.R.; Vuong, Q.; et al. OpenVLA: An Open-Source Vision-Language-Action Model. In Proceedings of the Conference on Robot Learning, 2025, Vol. 270, pp. 2679–2713, [2406.09246].
9. Chen, Q.; Kiami, S.C.; Gupta, A.; Kumar, V. GenAug: Retargeting behaviors to unseen situations via Generative Augmentation. In Proceedings of the Robotics: Science and Systems XIX, 2023, Vol. 19, [2302.06671]. https://doi.org/10.15607/RSS.2023.XIX.010.
10. Mandlekar, A.; Nasiriany, S.; Wen, B.; Akinola, I.; Narang, Y.; Fan, L.; Zhu, Y.; Fox, D. MimicGen: A Data Generation System for Scalable Robot Learning using Human Demonstrations. In Proceedings of the Conference on Robot Learning, 2023, Vol. 229, pp. 1820–1864, [2310.17596].
11. Wang, J.; Qin, Y.; Kuang, K.; Korkmaz, Y.; Gurumoorthy, A.; Su, H.; Wang, X. CyberDemo: Augmenting Simulated Human Demonstration for Real-World Dexterous Manipulation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 17952–17963, [2402.14795].
12. Haldar, S.; Pari, J.; Rai, A.; Pinto, L. Teach a Robot to FISH: Versatile Imitation from One Minute of Demonstrations. In Proceedings of the Robotics: Science and Systems XIX, 2023, Vol. 19, [2303.01497]. https://doi.org/10.15607/RSS.2023.XIX.009.
13. Luo, J.; Xu, C.; Wu, J.; Levine, S. Precise and dexterous robotic manipulation via human-in-the-loop reinforcement learning. *Science Robotics* **2025**, *10*, eads5033, [2410.21845]. https://doi.org/10.1126/scirobotics.ads5033.
14. Ross, S.; Gordon, G.; Bagnell, D. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning. In Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics, 2011, Vol. 15, pp. 627–635, [1011.0686].
15. Mandlekar, A.; Xu, D.; Wong, J.; Nasiriany, S.; Wang, C.; Kulkarni, R.; Li, F.F.; Savarese, S.; Zhu, Y.; Martín-Martín, R. What Matters in Learning from Offline Human Demonstrations for Robot Manipulation. In Proceedings of the Conference on Robot Learning, 2022, Vol. 164, pp. 1678–1690, [2108.03298].
16. Rajeswaran, A.; Kumar, V.; Gupta, A.; Vezzani, G.; Schulman, J.; Todorov, E.; Levine, S. Learning Complex Dexterous Manipulation with Deep Reinforcement Learning and Demonstrations. In Proceedings of the Robotics: Science and Systems XIV, 2018, Vol. 14, [1709.10087]. https://doi.org/10.15607/RSS.2018.XIV.049.
17. Jiang, R.; He, B.; Wang, Z.; Cheng, X.; Sang, H.; Zhou, Y. Robot skill learning and the data dilemma it faces: a systematic review. *Robotic Intelligence and Automation* **2024**, *44*, 270–286. https://doi.org/10.1108/ria-10-2023-0146.
18. Li, D.; Jin, Y.; Sun, Y.; A, Y.; Yu, H.; Shi, J.; Hao, X.; Hao, P.; Liu, H.; Li, X.; et al. What foundation models can bring for robot learning in manipulation: A survey. *The International Journal of Robotics Research* **2026**, *45*, 1091–1142, [2404.18201]. https://doi.org/10.1177/02783649251390579.

19. Ma, J.; Zhang, E.; Yang, H.; Li, D.; Xu, C.; Wang, G.; Wang, H. Robot Learning from Human Videos: A Survey. *arXiv* **2026**, [2604.27621]. Preprint, arXiv:2604.27621.
20. Wolf, R.; Shi, Y.; Liu, S.; Rayyes, R. Diffusion models for robotic manipulation: a survey. *Frontiers in Robotics and AI* **2025**, *12*, 1606247, [2504.08438]. https://doi.org/10.3389/frobt.2025.1606247.
21. Yu, Z.; Wang, B.; Zeng, P.; Zhang, H.; Zhang, J.; Wang, Z.; Gao, L.; Song, J.; Sebe, N.; Shen, H.T. A Survey on Efficient Vision-Language-Action Models. *arXiv* **2025**, [2510.24795]. Preprint, arXiv:2510.24795.
22. Guan, W.; Hu, Q.; Li, A.; Cheng, J. Efficient Vision-Language-Action Models for Embodied Manipulation: A Systematic Survey. *arXiv* **2025**, [2510.17111]. Preprint, arXiv:2510.17111.
23. Tang, C.; Abbatematteo, B.; Hu, J.; Chandra, R.; Martín-Martín, R.; Stone, P. Deep Reinforcement Learning for Robotics: A Survey of Real-World Successes. *Annual Review of Control, Robotics, and Autonomous Systems* **2025**, *8*, 153–188, [2408.03539]. https://doi.org/10.1146/annurev-control-030323-022510.
24. Xie, W.; Correll, N. Towards Forceful Robotic Foundation Models: a Literature Survey. *arXiv* **2025**, [2504.11827]. Preprint, arXiv:2504.11827.
25. Ma, Y.; Song, Z.; Zhuang, Y.; Hao, J.; King, I. A Survey on Vision–Language–Action Models for Embodied AI. *IEEE Transactions on Neural Networks and Learning Systems* **2026**, *37*, 3031–3051, [2405.14093]. https://doi.org/10.1109/tnnls.2025.3650584.
26. Ud Din, M.; Akram, W.; Saad Saoud, L.; Rosell, J.; Hussain, I. Multimodal fusion with vision-language-action models for robotic manipulation: A systematic review. *Information Fusion* **2026**, *129*, 104062. https://doi.org/10.1016/j.inffus.2025.104062.
27. Bai, S.; Song, W.; Chen, J.; Ji, Y.; Zhong, Z.; Yang, J.; Zhao, H.; Zhou, W.; Zhao, W.; Li, Z.; et al. Towards a Unified Understanding of Robot Manipulation: A Comprehensive Survey. *arXiv* **2025**, [2510.10903]. Preprint, arXiv:2510.10903.
28. Parnada, A.; Qu, M.; Castellani, M.; Chang, H.J.; Wang, Y. Towards cost-effective and safe contact-rich robotic manipulation with reinforcement learning: A review of techniques for future industrial automation. *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering* **2026**, *240*, 3–35. https://doi.org/10.1177/09596518251350353.
29. Acs, M.; Zhong, X. A scoping review of reinforcement learning methods for long horizon robotic manipulation. *Discover Robotics* **2026**, *2*, 22. https://doi.org/10.1007/s44430-026-00037-1.
30. Sarowar, M.S. What Matters: Datasets or Robust Frameworks in Modern Robot Learning? *Preprints.org* **2026**. Preprint, https://doi.org/10.20944/preprints202606.1149.v1.
31. Tsuji, T.; Kato, Y.; Solak, G.; Zhang, H.; Petrič, T.; Nori, F.; Ajoudani, A. A survey on imitation learning for contact-rich tasks in robotics. *The International Journal of Robotics Research* **2026**, p. 02783649261417694, [2506.13498]. https://doi.org/10.1177/02783649261417694.
32. Gao, J.; Belkhale, S.; Dasari, S.; Balakrishna, A.; Shah, D.; Sadigh, D. A Taxonomy for Evaluating Generalist Robot Manipulation Policies. *IEEE Robotics and Automation Letters* **2026**, *11*, 3182–3189, [2503.01238]. https://doi.org/10.1109/lra.2026.3656785.
33. Hou, B.; Li, G.; Jia, J.; An, T.; Guo, X.; Leng, S.; Geng, H.; Ze, Y.; Harada, T.; Torr, P.; et al. World Model for Robot Learning: A Comprehensive Survey. *arXiv* **2026**, [2605.00080]. Preprint, arXiv:2605.00080.
34. Black, K.; Brown, N.; Darpinian, J.; Dhabalia, K.; Driess, D.; Esmail, A.; Equi, M.R.; Finn, C.; Fusai, N.; Galliker, M.Y.; et al. $\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization. In Proceedings of the Conference on Robot Learning, 2025, Vol. 305, pp. 17–40, [2504.16054].
35. Fu, Z.; Zhao, T.Z.; Finn, C. Mobile ALOHA: Learning Bimanual Mobile Manipulation using Low-Cost Whole-Body Teleoperation. In Proceedings of the Conference on Robot Learning, 2025, Vol. 270, pp. 4066–4083, [2401.02117].
36. Shafiullah, N.M.M.; Rai, A.; Etukuru, H.; Liu, Y.; Misra, I.; Chintala, S.; Pinto, L. On Bringing Robots Home. *arXiv* **2023**, [2311.16098]. Preprint, arXiv:2311.16098.
37. Wu, P.; Shentu, Y.; Yi, Z.; Lin, X.; Abbeel, P. GELLO: A General, Low-Cost, and Intuitive Teleoperation Framework for Robot Manipulators. In Proceedings of the 2024 IEEE/RSJ International

Conference on Intelligent Robots and Systems (IROS), 2024, pp. 12156–12163, [2309.13037]. https://doi.org/10.1109/iros58592.2024.10801581.

38. Wang, C.; Shi, H.; Wang, W.; Zhang, R.; Li, F.F.; Liu, K. DexCap: Scalable and Portable Mocap Data Collection System for Dexterous Manipulation. In Proceedings of the Robotics: Science and Systems XX, 2024, [2403.07788]. https://doi.org/10.15607/rss.2024.xx.043.
39. Chen, S.; Wang, C.; Nguyen, K.; Li, F.F.; Liu, C.K. ARCap: Collecting High-Quality Human Demonstrations for Robot Learning with Augmented Reality Feedback. In Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA), 2025, pp. 8291–8298, [2410.08464]. https://doi.org/10.1109/icra55743.2025.11128717.
40. Xu, M.; Zhang, H.; Hou, Y.; Xu, Z.; Fan, L.; Veloso, M.; Song, S. DexUMI: Using Human Hand as the Universal Manipulation Interface for Dexterous Manipulation. In Proceedings of the Conference on Robot Learning, 2025, Vol. 305, pp. 437–459, [2505.21864].
41. Wu, K.; Hou, C.; Liu, J.; Che, Z.; Ju, X.; Yang, Z.; Li, M.; Zhao, Y.; Xu, Z.; Yang, G.; et al. RoboMIND: Benchmark on Multi-embodiment Intelligence Normative Data for Robot Manipulation. In Proceedings of the Robotics: Science and Systems XXI, 2025, Vol. 21, [2412.13877]. https://doi.org/10.15607/RSS.2025.XXI.152.
42. Gao, J.; Xie, A.; Xiao, T.; Finn, C.; Sadigh, D. Efficient Data Collection for Robotic Manipulation via Compositional Generalization. In Proceedings of the Robotics: Science and Systems XX, 2024, [2403.05110]. https://doi.org/10.15607/rss.2024.xx.013.
43. Fang, H.S.; Fang, H.; Tang, Z.; Liu, J.; Wang, C.; Wang, J.; Zhu, H.; Lu, C. RH20T: A Comprehensive Robotic Dataset for Learning Diverse Skills in One-Shot. In Proceedings of the 2024 IEEE International Conference on Robotics and Automation (ICRA), 2024, pp. 653–660, [2307.00595]. https://doi.org/10.1109/icra57147.2024.10611615.
44. AgiBot-World-Contributors.; Bu, Q.; Cai, J.; Chen, L.; Cui, X.; Ding, Y.; Feng, S.; Gao, S.; He, X.; Hu, X.; et al. AgiBot World Colosseo: A Large-Scale Manipulation Platform for Scalable and Intelligent Embodied Systems. In Proceedings of the 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025, pp. 3549–3556, [2503.06669]. https://doi.org/10.1109/iros60139.2025.11247088.
45. Li, H.; Cui, Y.; Sadigh, D. How to Train Your Robots? The Impact of Demonstration Modality on Imitation Learning. In Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA), 2025, pp. 1113–1120, [2503.07017]. https://doi.org/10.1109/icra55743.2025.11128520.
46. Hejna, J.; Bhateja, C.A.; Jiang, Y.; Pertsch, K.; Sadigh, D. ReMix: Optimizing Data Mixtures for Large Scale Imitation Learning. In Proceedings of the Conference on Robot Learning, 2025, Vol. 270, pp. 145–164, [2408.14037].
47. Hejna, J.; Mirchandani, S.; Balakrishna, A.; Xie, A.; Wahid, A.; Tompson, J.; Sanketi, P.R.; Shah, D.; Devin, C.M.; Sadigh, D. Robot Data Curation with Mutual Information Estimators. In Proceedings of the Robotics: Science and Systems XXI, 2025, Vol. 21, [2502.08623]. https://doi.org/10.15607/RSS.2025.XXI.023.
48. Parekh, S.; Nemlekar, H.; Losey, D.P. Towards balanced behavior cloning from imbalanced datasets. *Autonomous Robots* **2026**, *50*, 9, [2508.06319]. https://doi.org/10.1007/s10514-025-10237-0.
49. Liu, H.; Nasiriany, S.; Zhang, L.; Bao, Z.; Zhu, Y. Robot learning on the job: Human-in-the-loop autonomy and learning during deployment. *The International Journal of Robotics Research* **2025**, *44*, 1727–1742, [2211.08416]. https://doi.org/10.1177/02783649241273901.
50. Hoque, R.; Balakrishna, A.; Novoseller, E.; Wilcox, A.; Brown, D.S.; Goldberg, K. ThriftyDAgger: Budget-Aware Novelty and Risk Gating for Interactive Imitation Learning. In Proceedings of the Conference on Robot Learning, 2022, Vol. 164, pp. 598–608, [2109.08273].
51. Bharadhwaj, H.; Vakil, J.; Sharma, M.; Gupta, A.; Tulsiani, S.; Kumar, V. RoboAgent: Generalization and Efficiency in Robot Manipulation via Semantic Augmentations and Action Chunking. In Proceedings of the 2024 IEEE International Conference on Robotics and Automation (ICRA), 2024, pp. 4788–4795, [2309.01918]. https://doi.org/10.1109/icra57147.2024.10611293.
52. Jiang, Z.; Xie, Y.; Lin, K.; Xu, Z.; Wan, W.; Mandlekar, A.; Fan, L.J.; Zhu, Y. DexMimicGen: Automated Data Generation for Bimanual Dexterous Manipulation via Imitation Learning. In

Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA), 2025, pp. 16923–16930, [2410.24185]. https://doi.org/10.1109/icra55743.2025.11127809.

53. Hoque, R.; Mandlekar, A.; Garrett, C.; Goldberg, K.; Fox, D. IntervenGen: Interventional Data Generation for Robust and Data-Efficient Robot Imitation Learning. In Proceedings of the 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024, pp. 2840–2846, [2405.01472]. https://doi.org/10.1109/iros58592.2024.10801523.
54. Xue, Z.; Deng, S.; Chen, Z.; Wang, Y.; Yuan, Z.; Xu, H. DemoGen: Synthetic Demonstration Generation for Data-Efficient Visuomotor Policy Learning. In Proceedings of the Robotics: Science and Systems XXI, 2025, [2502.16932]. https://doi.org/10.15607/rss.2025.xxi.157.
55. Ha, H.; Florence, P.; Song, S. Scaling Up and Distilling Down: Language-Guided Robot Skill Acquisition. In Proceedings of the Conference on Robot Learning, 2023, Vol. 229, pp. 3766–3777, [2307.14535].
56. Nasiriany, S.; Maddukuri, A.; Zhang, L.; Parikh, A.; Lo, A.; Joshi, A.; Mandlekar, A.; Zhu, Y. RoboCasa: Large-Scale Simulation of Household Tasks for Generalist Robots. In Proceedings of the Robotics: Science and Systems XX, 2024, Vol. 20, [2406.02523]. https://doi.org/10.15607/RSS.2024.XX.050.
57. Chen, T.; Chen, Z.; Chen, B.; Cai, Z.; Liu, Y.; Li, Z.; Liang, Q.; Lin, X.; Ge, Y.; Gu, Z.; et al. RoboTwin 2.0: A Scalable Data Generator and Benchmark with Strong Domain Randomization for Robust Bimanual Robotic Manipulation. *arXiv* **2025**, [2506.18088]. arXiv technical version; paper listed in ICML 2026 program.
58. Ji, G.; Polavaram, H.; Chen, L.Y.; Bajamahal, S.; Ma, Z.; Adebola, S.; Xu, C.; Goldberg, K. OXE-AugE: A Large-Scale Robot Augmentation of OXE for Scaling Cross-Embodiment Policy Learning. *arXiv* **2025**, [2512.13100]. Preprint, arXiv:2512.13100.
59. Oh, H.; Murooka, M.; Motoda, T.; Nakajo, R.; Domae, Y. Self-augmented robot trajectory: efficient imitation learning via safe self-augmentation with demonstrator-annotated precision. *Advanced Robotics* **2026**, pp. 1–15, [2509.09893]. https://doi.org/10.1080/01691864.2026.2631642.
60. Liao, H.; Wang, P.; Chen, S.; Cheng, K.; Chen, X.; Lin, Y.; Lin, M.; Liang, Z.; Fan, X.; Xing, C.; et al. DynamicManip: Enabling Dynamic Manipulation from a Single Static Demonstration. *arXiv* **2026**, [2608.01452]. Preprint, arXiv:2608.01452.
61. Zhao, M.; Huang, D.; Tang, Y.; Li, P.; Yan, M.; Zhuang, R.; Huang, Y.; Wang, J.; Zhai, H.; Zhou, T.; et al. AXIS: A Growable Community-Driven Data Engine for Scalable Robot Manipulation. *arXiv* **2026**, [2607.21588]. Preprint, arXiv:2607.21588.
62. Gu, C.; Kang, H.; Lin, J.; Wang, J.; Wu, D.; Xie, S.; Huang, F.; Ge, J.; Gong, Z.; Li, L.; et al. IGen: Scalable Data Generation for Robot Learning from Open-World Images. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026, pp. 28111–28122, [2512.01773].
63. Shridhar, M.; Manuelli, L.; Fox, D. Perceiver-Actor: A Multi-Task Transformer for Robotic Manipulation. In Proceedings of the Conference on Robot Learning, 2023, Vol. 205, pp. 785–799, [2209.05451].
64. Goyal, A.; Xu, J.; Guo, Y.; Blukis, V.; Chao, Y.W.; Fox, D. RVT: Robotic View Transformer for 3D Object Manipulation. In Proceedings of the Conference on Robot Learning, 2023, Vol. 229, pp. 694–710, [2306.14896].
65. Ze, Y.; Zhang, G.; Zhang, K.; Hu, C.; Wang, M.; Xu, H. 3D Diffusion Policy: Generalizable Visuomotor Policy Learning via Simple 3D Representations. In Proceedings of the Robotics: Science and Systems XX, 2024, Vol. 20, [2403.03954]. https://doi.org/10.15607/RSS.2024.XX.067.
66. Belkhale, S.; Cui, Y.; Sadigh, D. HYDRA: Hybrid Robot Actions for Imitation Learning. In Proceedings of the Conference on Robot Learning, 2023, Vol. 229, pp. 2113–2133, [2306.17237].
67. Wang, C.; Fang, H.; Fang, H.S.; Lu, C. RISE: 3D Perception Makes Real-World Robot Imitation Simple and Effective. In Proceedings of the 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024, pp. 2870–2877, [2404.12281]. https://doi.org/10.1109/iros58592.2024.10801678.

68. Prasad, A.; Lin, K.; Wu, J.; Zhou, L.; Bohg, J. Consistency Policy: Accelerated Visuomotor Policies via Consistency Distillation. In Proceedings of the Robotics: Science and Systems XX, 2024, [2405.07503]. https://doi.org/10.15607/rss.2024.xx.071.
69. Lee, S.; Wang, Y.; Etukuru, H.; Kim, H.J.; Shafiullah, N.M.M.; Pinto, L. Behavior Generation with Latent Actions. In Proceedings of the International Conference on Machine Learning, 2024, Vol. 235, pp. 26991–27008, [2403.03181].
70. Nair, S.; Rajeswaran, A.; Kumar, V.; Finn, C.; Gupta, A. R3M: A Universal Visual Representation for Robot Manipulation. In Proceedings of the Conference on Robot Learning, 2023, Vol. 205, pp. 892–909, [2203.12601].
71. Majumdar, A.; Yadav, K.; Arnaud, S.; Ma, J.; Chen, C.; Silwal, S.; Jain, A.; Berges, V.P.; Wu, T.; Vakil, J.; et al. Where are we in the search for an Artificial Visual Cortex for Embodied Intelligence? In Proceedings of the Advances in Neural Information Processing Systems, 2023, Vol. 36, pp. 655–677, [2303.18240]. https://doi.org/10.52202/075280-0031.
72. Brohan, A.; Brown, N.; Carbajal, J.; Chebotar, Y.; Dabis, J.; Finn, C.; Gopalakrishnan, K.; Hausman, K.; Herzog, A.; Hsu, J.; et al. RT-1: Robotics Transformer for Real-World Control at Scale. In Proceedings of the Robotics: Science and Systems XIX, 2023, Vol. 19, [2212.06817]. https://doi.org/10.15607/RSS.2023.XIX.025.
73. Zitkovich, B.; Yu, T.; Xu, S.; Xu, P.; Xiao, T.; Xia, F.; Wu, J.; Wohlhart, P.; Welker, S.; Wahid, A.; et al. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. In Proceedings of the Conference on Robot Learning, 2023, Vol. 229, pp. 2165–2183, [2307.15818].
74. Kim, M.J.; Finn, C.; Liang, P. Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success. In Proceedings of the Robotics: Science and Systems XXI, 2025, Vol. 21, [2502.19645]. https://doi.org/10.15607/RSS.2025.XXI.017.
75. Black, K.; Brown, N.; Driess, D.; Esmail, A.; Equi, M.R.; Finn, C.; Fusai, N.; Groom, L.; Hausman, K.; Ichter, B.; et al. π_0: A Vision-Language-Action Flow Model for General Robot Control. In Proceedings of the Robotics: Science and Systems XXI, 2025, Vol. 21, [2410.24164]. https://doi.org/10.15607/RSS.2025.XXI.010.
76. NVIDIA.; Bjorck, J.; Castañeda, F.; Cherniadev, N.; Da, X.; Ding, R.; Fan, L.J.; Fang, Y.; Fox, D.; Hu, F.; et al. GR00T N1: An Open Foundation Model for Generalist Humanoid Robots. *arXiv* **2025**, [2503.14734]. Preprint, arXiv:2503.14734.
77. Pertsch, K.; Stachowicz, K.; Ichter, B.; Driess, D.; Nair, S.; Vuong, Q.; Mees, O.; Finn, C.; Levine, S. FAST: Efficient Action Tokenization for Vision-Language-Action Models. In Proceedings of the Robotics: Science and Systems XXI, 2025, [2501.09747]. https://doi.org/10.15607/rss.2025.xxi.012.
78. Ankile, L.; Jiang, Z.; Duan, R.; Shi, G.; Abbeel, P.; Nagabandi, A. Residual Off-Policy RL for Finetuning Behavior Cloning Policies. *arXiv* **2025**, [2509.19301]. arXiv technical version; also presented at ICLR 2026 Lifelong Agent Workshop.
79. Ren, A.; Lidard, J.; Ankile, L.; Simeonov, A.; Agrawal, P.; Majumdar, A.; Burchfiel, B.; Dai, H.; Simchowitz, M. Diffusion Policy Policy Optimization. In Proceedings of the International Conference on Learning Representations, 2025, Vol. 2025, pp. 77288–77329, [2409.00588].
80. Shukor, M.; Aubakirova, D.; Capuano, F.; Kooijmans, P.; Palma, S.; Zouitine, A.; Aractingi, M.; Pascal, C.; Russi, M.; Marafioti, A.; et al. SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics. *arXiv* **2025**, [2506.01844]. Preprint, arXiv:2506.01844.
81. Wang, C.; Fan, L.; Sun, J.; Zhang, R.; Li, F.F.; Xu, D.; Zhu, Y.; Anandkumar, A. MimicPlay: Long-Horizon Imitation Learning by Watching Human Play. In Proceedings of the Conference on Robot Learning, 2023, Vol. 229, pp. 201–221, [2302.12422].
82. Ball, P.J.; Smith, L.; Kostrikov, I.; Levine, S. Efficient Online Reinforcement Learning with Offline Data. In Proceedings of the International Conference on Machine Learning, 2023, Vol. 202, pp. 1577–1594, [2302.02948].
83. Laskey, M.; Lee, J.; Fox, R.; Dragan, A.; Goldberg, K. DART: Noise Injection for Robust Imitation Learning. In Proceedings of the Conference on Robot Learning, 2017, Vol. 78, pp. 143–156, [1703.09327].
84. Yang, J.; Mark, M.S.; Vu, B.; Sharma, A.; Bohg, J.; Finn, C. Robot Fine-Tuning Made Easy: Pre-Training Rewards and Policies for Autonomous Real-World Reinforcement Learning. In

Proceedings of the 2024 IEEE International Conference on Robotics and Automation (ICRA), 2024, pp. 4804–4811, [2310.15145]. https://doi.org/10.1109/icra57147.2024.10610421.

85. Liu, B.; Zhu, Y.; Gao, C.; Feng, Y.; Liu, Q.; Zhu, Y.; Stone, P. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. In Proceedings of the Advances in Neural Information Processing Systems, 2023, Vol. 36, pp. 44776–44791, [2306.03310]. https://doi.org/10.52202/075280-1939.
86. Pumacay, W.; Singh, I.; Duan, J.; Krishna, R.; Thomason, J.; Fox, D. THE COLOSSEUM: A Benchmark for Evaluating Generalization for Robotic Manipulation. In Proceedings of the Robotics: Science and Systems XX, 2024, Vol. 20, [2402.08191]. https://doi.org/10.15607/RSS.2024.XX.133.
87. Li, X.; Hsu, K.; Gu, J.; Mees, O.; Pertsch, K.; Walke, H.R.; Fu, C.; Lunawat, I.; Sieh, I.; Kirmani, S.; et al. Evaluating Real-World Robot Manipulation Policies in Simulation. In Proceedings of the Conference on Robot Learning, 2025, Vol. 270, pp. 3705–3728, [2405.05941].
88. Jiang, T.; Tan, X.; Wheeler, S.; Sun, L.; Ayalew, T.W.; Walter, M. What Are We Actually Benchmarking in Robot Manipulation? *arXiv* **2026**, [2606.04233]. Preprint, arXiv:2606.04233.